

Abhinav Basam

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OBJECTIVE

Motivated AI & ML undergraduate seeking a software/ML internship to apply hands-on skills in machine learning, deep learning, and data science. Experienced in building end-to-end ML pipelines, implementing core algorithms from scratch, and deploying computer vision and NLP models. Eager to contribute to teams that work on impactful, data-driven products.

EDUCATION

B V Raju Institute of Technology (BVRIT)

B.Tech – Artificial Intelligence & Machine Learning

Sri Chaitanya Junior College

Intermediate (MPC) – 94.3%

Narsapur, Telangana

2024 – 2028

Hyderabad, Telangana

2022 – 2024

RELEVANT COURSEWORK

Data Structures & Algorithms | Linear Algebra & Calculus | Probability & Statistics | Object-Oriented Programming | Database Management Systems | Machine Learning Fundamentals | Computer Vision

PROJECTS

ML Algorithms from Scratch | *Python, NumPy, Matplotlib*

Jan 2026 – Present

- Implemented Linear Regression, Logistic Regression, K-Nearest Neighbours, and K-Means Clustering using only NumPy, without any ML library abstractions, to build deep algorithmic understanding
- Built gradient descent optimisation from scratch, including learning-rate tuning and convergence diagnostics, achieving <0.5% deviation from scikit-learn's output on benchmark datasets
- Validated all implementations against scikit-learn baselines and visualised decision boundaries, loss curves, and cluster assignments with Matplotlib to confirm correctness

Customer Churn Prediction – End-to-End ML Pipeline | *Python, scikit-learn, pandas, Flask*

Nov 2025 – Jan 2026

- Developed a full ML pipeline on the Telco Customer Churn dataset (7,000+ records), covering EDA, feature engineering, and handling class imbalance via SMOTE, achieving an F1-score of 0.83
- Compared Random Forest, XGBoost, and Logistic Regression using cross-validation; selected the best model based on ROC-AUC and precision-recall trade-offs on the imbalanced test set
- Packaged the trained model as a REST API using Flask and Docker, enabling batch inference via HTTP endpoints; deployed locally and tested with Postman

Jarvis – Offline AI Voice Assistant | *Python, Speech Recognition, OS Automation*

Jan 2026 – Present

- Built an offline AI voice assistant pipeline with wake-word detection, speech recognition, and command execution, reducing end-to-end response latency by ~30% vs. cloud-based baselines
- Implemented utterance-level RMS energy analysis to accurately detect speech boundaries, improving transcription accuracy in noisy environments
- Designed a finite-state listening system to reliably manage wake, active-listening, and idle states, with automated OS-level action dispatch

Dog Breed Classification using CNN | *Python, TensorFlow/Keras, Deep Learning*

Sept 2025 – Dec 2025

- Built a CNN image classifier to identify 10 dog breeds, achieving 89% validation accuracy on a 5,000-image dataset after hyperparameter tuning
- Applied data augmentation (random flips, rotations, zoom) and transfer learning with MobileNetV2 to improve generalisation on limited training data
- Analysed training vs. validation loss/accuracy curves to diagnose and resolve overfitting, reducing the generalisation gap from 14% to 4%

TECHNICAL SKILLS

Languages: Python, Java, C, SQL

Machine Learning: Supervised & Unsupervised Learning, CNNs, Transfer Learning, Bias-Variance Tradeoff, Regularisation (L1/L2), Feature Engineering, Class Imbalance (SMOTE), Cross-Validation

Frameworks & Libraries: NumPy, pandas, scikit-learn, TensorFlow, Keras, XGBoost, Matplotlib, Seaborn, Flask, FastAPI, React

Tools & Platforms: Git, Docker, Jupyter Notebook, VS Code, GCP, Postman

Core CS: Data Structures & Algorithms, DBMS, OOP, REST APIs

CERTIFICATIONS

NPTEL Programming, Data Structures and Algorithms using Python	July 2025 – Sept 2025
Infosys Springboard Basics of Python	May 2025 – June 2025

ACHIEVEMENTS & ACTIVITIES

NPTEL Elite Badge Scored in top 25% nationally in the DSA using Python course (IIT Madras)	Sept 2025
Kaggle Notebooks Published reproducible EDA and ML pipeline notebooks; accumulated 50+ views	2025 – Present
Open Source Maintaining ML-from-Scratch repository on GitHub documenting algorithm implementations with tests	2026 – Present
Intermediate Board Secured 94.3% in MPC stream, Telangana State Board	2024